

Tianlong Wang

PhD candidate | Zhejiang University | Geohazards, hydrology and artificial intelligence

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Research profile

I develop physics-informed and explainable machine-learning methods for debris-flow, landslide and hydrological prediction. My research combines hydrological reasoning, residual learning, ensemble optimization and uncertainty analysis, with an emphasis on small catchments and sparse observations. I expect to complete my PhD in June 2027 and seek international postdoctoral opportunities from July 2027 in hydrology, geohazards, geotechnics and related AI applications.

Research outputs: 19 published papers (18 journal articles and 1 conference paper); 5 manuscripts under review; 3 investigator-led research awards totalling RMB 126,000; 6 open-source research software projects.

Education and joint doctoral training

Zhejiang University | PhD candidate | Sep 2022 - Jun 2027 (expected)

Ocean College; Ocean Technology and Engineering; Supervisor: [Hongyue Sun](#)

Nanyang Technological University | Joint PhD Student | Sep 2024 - Sep 2026

School of Civil and Environmental Engineering; China Scholarship Council sponsorship; Supervisor: [Chu Jian](#)

Research projects and funding

Investigator-led research awards

1. **Spatiotemporal Dynamic Forecasting and Interpretive Analysis System for Debris Flows in Small Watersheds.** Young Elite Scientists Sponsorship Program Doctoral Student Special Plan, China Association for Science and Technology; PI; January 2025 — December 2026. Project ID: 156-O-170-0000603-2 Funding: RMB 40,000.
2. **Research on Intelligent Debris Flow Forecasting Methods for Small Watersheds Based on Multi-scale Hydrodynamic Coupling Mechanisms.** Seed Fund Cultivation Project of Ocean College, Zhejiang University; PI; June 2025 — June 2027. Grant No. 2025BS004 Funding: RMB 50,000.
3. **Spatiotemporal Dynamic Forecasting and Interpretive Analysis System for Debris Flows in Small Watersheds.** Outstanding Doctoral Dissertation Award Cultivation Grant, Zhejiang University; PI; 2025–2026. Grant No. 825011A · RMB 36,000.

Collaborative research projects

1. **Early Forecasting and Disaster Control Methods for Debris Flows in Small Watersheds Based on Hydrodynamic Analysis.** National Natural Science Foundation of China, Key Program; Core team member; January 2023 — December 2027. Grant No. 42230702.
2. **Research on Deformation Prediction Models and Instability Criterion Methods for Reservoir Bank Slopes Based on Copula Theory.** National Natural Science Foundation of China, Young Scientists Fund; Core team member; January 2019 — December 2021. Grant No. 51809151. Completed.

Manuscripts under review

1. **Tianlong Wang**, Jian Chu, Hao Yang, Hongyue Sun*. Physics-guided residual stacking for short-lead runoff and stage forecasting in the Fengshuwan catchment, China. *Journal of Hydrology: Regional Studies*. Under review following revision.
2. **Tianlong Wang**, Jian Chu, Hao Yang, Hongyue Sun*. Basin Water Storage Degree for Risk-Informed Debris-Flow Warning Support: A Forward-Looking Proof of Concept in a Small Mountain Catchment. *International Journal of Disaster Risk Reduction*. Under review following revision.

3. **Tianlong Wang**, Jian Chu, Jingkun Su, Runting Fan, Hao Yang, Hongyue Sun*. Hydrological readiness forecasting for debris flow early warning in a mountain catchment with sparse event records. *Bulletin of Engineering Geology and the Environment*. Under review.
4. **Tianlong Wang**, Jian Chu, Hao Yang, Hongyue Sun*. Robust state estimation and hydrology-informed residual learning for landslide velocity forecasting with prequential prediction intervals. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards*. Under review.
5. **Tianlong Wang**, Jian Chu, Yanhui Li, Hao Yang, Hongyue Sun*. Sediment-source information improves small-sample prediction of maximum recorded debris-flow volume with a pretrained tabular model. *Stochastic Environmental Research and Risk Assessment*. Under review following revision.

Published papers

1. **Tianlong Wang***, H. Yang. DOME: A Dynamic Optimization Meta-Ensemble Framework for Debris-Flow Susceptibility Assessment with SHAP-Based Interpretation. *Natural Hazards*, 122, Article 545, 2026. <https://doi.org/10.1007/s11069-026-08307-2>
2. **Tianlong Wang**, H. Yang, H. Sun*. Intelligent Prediction of Post-Seismic Debris Flow Source Volumes: A Physics-Informed Generative Deep Learning Framework. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards*, 1–29, 2026. <https://doi.org/10.1080/17499518.2026.2651228>
3. K. Zhang, **Tianlong Wang**, J. Su, H. Sun*. Impacts of Land Use Change on Watershed Hydrological Processes Based on the SWAT-ML-SHAP Model: A Case Study of Fenshuijiang River Basin. *Earth Science*, 2026. <https://doi.org/10.3799/dqkx.2025.290>
4. H. Yang, **Tianlong Wang***, N. I. Fomin, S. Xiao, L. Liu. An integrated IKOA-CNN-BiGRU-Attention framework with SHAP explainability for high-precision debris flow hazard prediction in the Nuijiang river basin, China. *PLOS ONE*, 20(6), e0326587, 2025. <https://doi.org/10.1371/journal.pone.0326587>
5. **Tianlong Wang**, Q. Ge, T. Ma, et al.. A novel method for predicting debris flow hazard: a multi-strategy fusion approach based on the light gradient boosting machine framework. *Stochastic Environmental Research and Risk Assessment*, 39, 4867–4890, 2025. <https://doi.org/10.1007/s00477-025-02955-9>
6. X. Wang, Q. Ge, K. Zhang, **Tianlong Wang**, T. Ma, H. Sun*. 3D Numerical Simulation of the Shortest Length of a Blocking Section for a Self-Starting Drainage Method. *International Journal of Geomechanics*, 25(10), 04025209, 2025. <https://doi.org/10.1061/IJGNALGMENG-10939>
7. P. Lu, Y. Yu*, S. Liu, **Tianlong Wang**, Z. Xu. Multi-parameter Calibration Method and Application of Constitutive Model Based on Improved Dung Beetle Optimization Algorithm. *Journal of Disaster Prevention and Mitigation Engineering*, 45(5), 1014–1023. (In Chinese), 2025. <https://doi.org/10.13409/j.cnki.jdpme.20250224001>
8. **Tianlong Wang**, K. Zhang, Z. Liu, T. Ma, R. Luo, H. Chen, X. Wang, W. Ge, H. Sun*. Prediction and explanation of debris flow velocity based on multi-strategy fusion Stacking ensemble learning model. *Journal of Hydrology*, 638, 131347, 2024. <https://doi.org/10.1016/j.jhydrol.2024.131347>
9. **Tianlong Wang**, R. Luo, T. Ma, et al.. Study and verification on an improved comprehensive prediction model of landslide displacement. *Bulletin of Engineering Geology and the Environment*, 83, Article 90, 2024. <https://doi.org/10.1007/s10064-024-03581-5>
10. Z. Chu, S. Yu, X. Li, X. Lu, X. Wang, R. Luo, **Tianlong Wang***. Indoor Model Test of Negative Pressure Drainage Method for Treatment of Engineering Mud. *Journal of Changjiang River Scientific Research Institute*, 41(9), 114–122, 129. (In Chinese), 2024. <https://doi.org/10.11988/ckyyb.20230665>
11. **Tianlong Wang**, D. Peng, X. Wang, B. Wu, R. Luo, Z. Chu, H. Sun*. Study on wavelet multi-scale analysis and prediction of landslide groundwater. *Journal of Hydroinformatics*, 26(1), 237–254, 2024. <https://doi.org/10.2166/hydro.2023.299>
12. H. Sun, X. Wang, Y. Yu*, J. Shu, **Tianlong Wang**, R. Luo, Z. Chu. A new anchor-siphon drainage combined method used for slope stabilization. *Journal of Mountain Science*, 20(11), 3295–3311, 2023. <https://doi.org/10.1007/s11629-023-7969-x>
13. Zheng Li, Jie Liu*, Qian Shi, Kun Mao, Zhao Li, Hongyu Tang, **Tianlong Wang**. Experimental Study on Grouting and Flow Characteristics of Rough Vertical Fractures. *Geofluids*, 2022, Article 3165524, 15 pages, 2022. <https://doi.org/10.1155/2022/3165524>

14. X. Xu, X. Lei, **Tianlong Wang**, G. Kong*, C. Li. Experimental Research on Shear Mechanical Characteristics of Pile-Coral Sand Interface. *Chinese Journal of Underground Space and Engineering*, 18(6), 1891–1897. (In Chinese), 2022. <https://doi.org/10.20174/j.juse.2022.06.016>
15. **Tianlong Wang**, Xiaorui Tao, Dong-kun Wu, Haotian Feng. Determining the location of River Dam Group based on set cover Model: A case study of Zambezi River Basin. *Scientific Research and Reviews*, 14, 122, 2021. <https://doi.org/10.28933/srr-2021-03-1005>
16. **Tianlong Wang**, J. Wang, X. Shao. Carbon emission forecast of China Transport Industry Based on Grey - Markov. *International Conference on Computing and Big Data (ICCBD)*, 2021. <https://doi.org/10.1109/ICCBD53833.2021.00192>
17. **Tianlong Wang**, H. Zhang, X. Tao, W. Chu. Research on Prediction and Optimization of Slope Deformation in Mining Area. *Environment, Resource and Ecology Journal*, 4, 66–69, 2020. <https://doi.org/10.23977/erej.2020.040110>
18. **Tianlong Wang**, Hongyan Bao, Hairui Zhang. Optimization of Task Pricing Based on Multiple Regression Analysis and Game Theory. *Journal of eSciences*, 3, 9, 2020. <https://doi.org/10.28933/esciences-2020-01-1805>
19. **Tianlong Wang**, Xiaorui Tao, Jiamei Zhang, Yulei Li. Predicting Traffic Congestion Time Based on Kalman Filter Algorithm. *Advances in Research and Reviews*, 1, 7, 2020. <https://doi.org/10.28933/arr-2020-06-2205>

Open-source research software

1. **DOMe**. Dynamic ensemble selection and optimization for regional debris-flow susceptibility, with SHAP interpretation. [Repository](#)
2. **TabPFNv2.5-DebrisFlow**. Small-sample debris-flow volume prediction using a pretrained tabular model and sediment-source information. [Repository](#)
3. **RSRL-HU**. One-day-ahead landslide velocity forecasts with hydrology-informed residual learning and sequential intervals. [Repository](#)
4. **Hydrological Readiness Forecasting**. Three-day probabilistic forecasts of warning-relevant hydrological states, evaluated chronologically. [Repository](#)
5. **BWSD**. Water-balance-based catchment storage indicator and daily hydrological-state classification. [Repository](#)
6. **HydroPIML-RS**. Physics-guided residual stacking for short-lead runoff and water-stage prediction. [Repository](#)

Academic service

Journal reviewer (16 journals): Journal of Hydrology; Acta Geotechnica; Bulletin of Engineering Geology and the Environment; Stochastic Environmental Research and Risk Assessment; Powder Technology; PLOS ONE; Applied Intelligence; Earth Science Informatics; Groundwater for Sustainable Development; International Journal of Computational Intelligence Systems; Journal of Rock Mechanics and Geotechnical Engineering; Air, Soil and Water Research; Discover Geoscience; Earth Sciences; Array; International Journal of Interactive Multimedia and Artificial Intelligence.

Memberships: Chinese Society for Rock Mechanics and Engineering; Geological Society of China; China Civil Engineering Society.

Honors and awards

1. **National Scholarship**. China; 2025, 2021.
2. **Outstanding Graduate**. Chinese Society for Rock Mechanics and Engineering · One of 100 recipients nationwide; 2022.
3. **Silver Award**. China International College Students' Innovation Competition, Singapore Regional; 2025.
4. **Grand Prize**. The 3rd National College Student Structural Design Information Technology Competition, China; 2021.